

Computer Organisation and Architecture

Assignment I - Semester I

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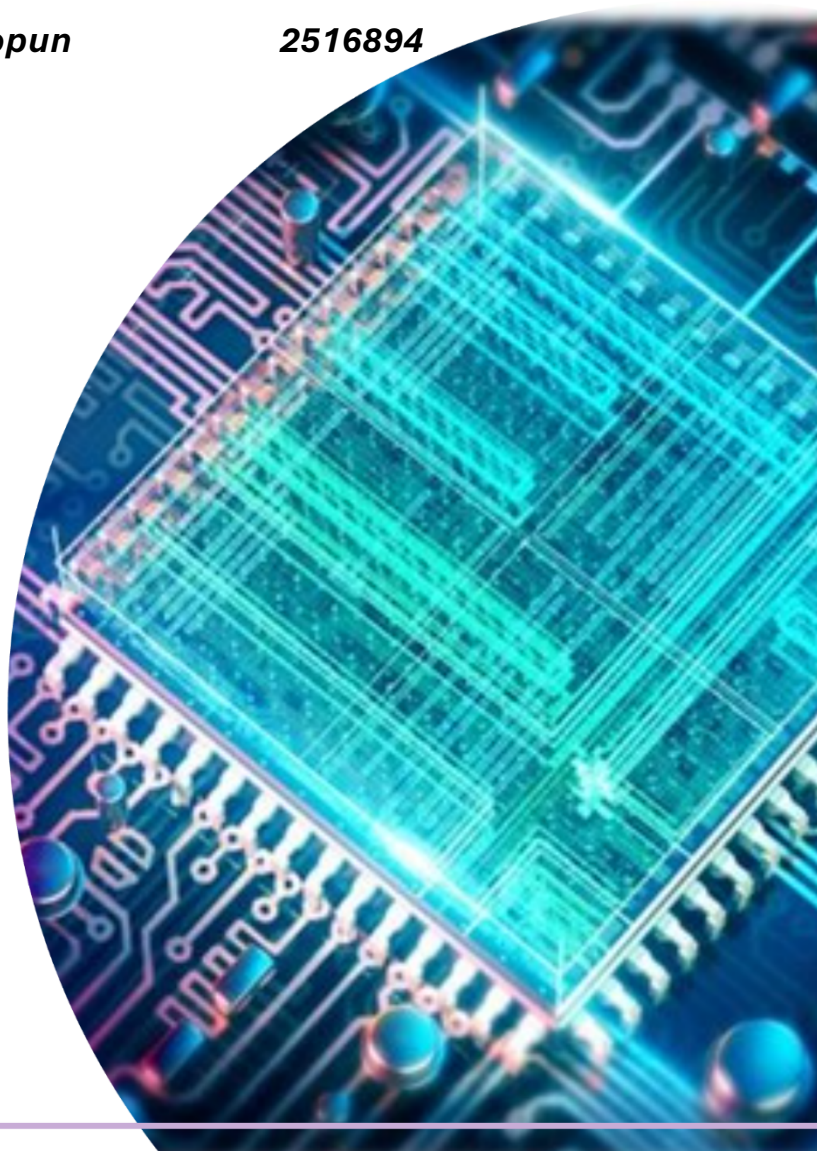


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Question 1:

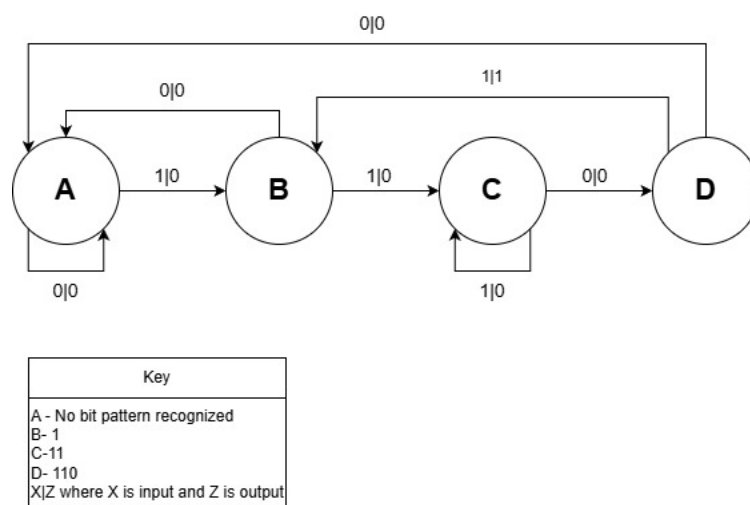
Implement a circuit that recognizes the occurrence of the sequence of bits 1101 on X by making Z equal to 1 when the previous three inputs to the circuit were 110 and current input is a 1. Use D flip-flops.

For example:

X: 0 0 1 1 0 0 1 1 0 1 1 0 1 0 0 1 1 1 ...

Y: 0 0 0 0 0 0 0 0 0 1 0 0 1 0 0 0 0 0

1.1 State Transition Diagram



1.1.1 State table

Present State	Input	Next State	Output
A	0	A	0
A	1	B	0
B	0	A	0
B	1	C	0
C	0	D	0
C	1	C	0
D	0	A	0
D	1	B	1

State	Binary Code
A	00
B	01
C	10
D	11

1.2 Truth Table

Present State $Q_1 \quad Q_0$		Input X	Next State $Q_1 \quad Q_0$		Flip-flop Input $D_1 \quad D_0$	Output Z
0	0	0	0	0	0 0	0
0	0	1	0	1	0 1	0
0	1	0	0	0	0 0	0
0	1	1	1	0	1 0	0
1	0	0	1	1	1 1	0
1	0	1	1	0	1 0	0
1	1	0	0	0	0 0	0
1	1	1	0	1	0 1	1

Next state is the same as Flip flop inputs

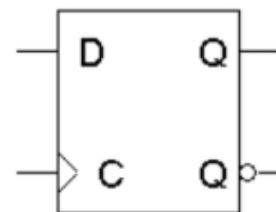
$$D_1 = Q_1' \cdot Q_0 \cdot X + Q_1 \cdot Q_0' \cdot X' + Q_1 \cdot Q_0' \cdot X$$

$$D_0 = Q_1' \cdot Q_0' \cdot X + Q_1 \cdot Q_0' \cdot X' + Q_1 \cdot Q_0 \cdot X$$

$$Z = Q_1 \cdot Q_0 \cdot X$$

1.2.1 Excitation table for D-flip flop

$Q(t)$	$Q(t+1)$	D	Operation
0	0	0	Reset
0	1	1	Set
1	0	0	Reset
1	1	1	Set



1.3 Karnaugh Map

1.3.1 K-Map for D_1

$X \backslash Q_1 Q_0$		$Q_1 Q_0$			
		$\bar{Q}_1 \bar{Q}_0$	$\bar{Q}_1 Q_0$	$Q_1 Q_0$	$Q_1 \bar{Q}_0$
$\bar{X}$	0	0	0	0	1
X	1	1	1	0	1

$$D_1 = Q_1 \cdot \bar{Q}_0 + \bar{Q}_1 \cdot Q_0 \cdot X$$

1.3.2 K-Map for D_0

$X \backslash Q_1 Q_0$		$Q_1 Q_0$			
		$\bar{Q}_1 \bar{Q}_0$	$\bar{Q}_1 Q_0$	$Q_1 Q_0$	$Q_1 \bar{Q}_0$
$\bar{X}$	0	0	0	0	1
X	1	1	0	1	0

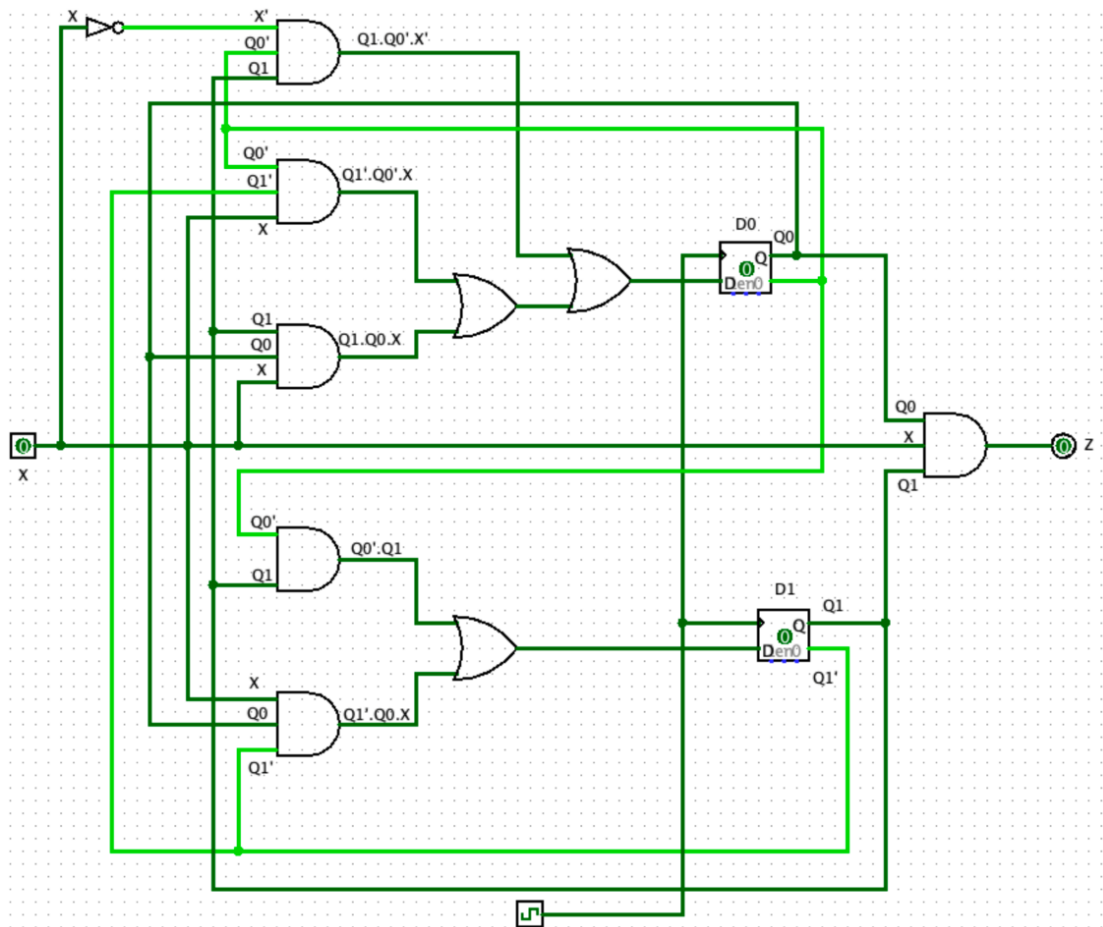
$$D_0 = \bar{Q}_1 \cdot \bar{Q}_0 \cdot X + Q_1 \cdot Q_0 \cdot X + Q_1 \cdot \bar{Q}_0 \cdot \bar{X}$$

1.3.2 K-Map for Z

$X \backslash Q_1 Q_0$		$Q_1 Q_0$			
		$\bar{Q}_1 \bar{Q}_0$	$\bar{Q}_1 Q_0$	$Q_1 Q_0$	$Q_1 \bar{Q}_0$
$\bar{X}$	0	0	0	0	0
X	1	0	0	1	0

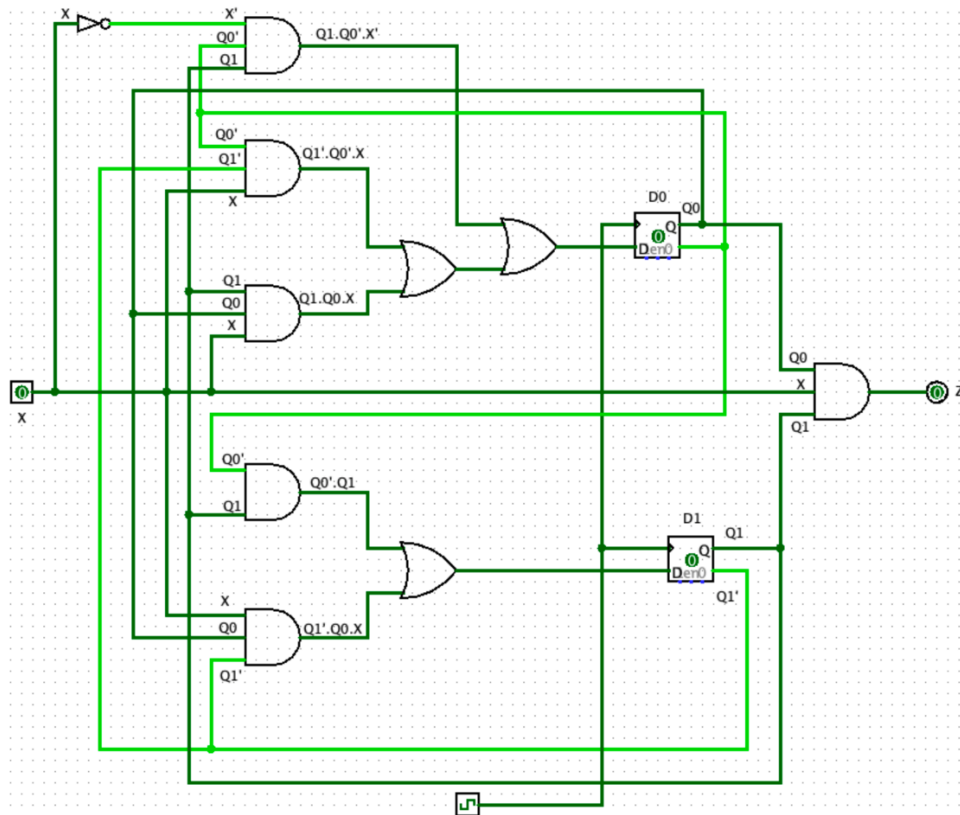
$$Z = Q_1 \cdot Q_0 \cdot X$$

1.4 Main Circuit Diagram (LOGISIM)

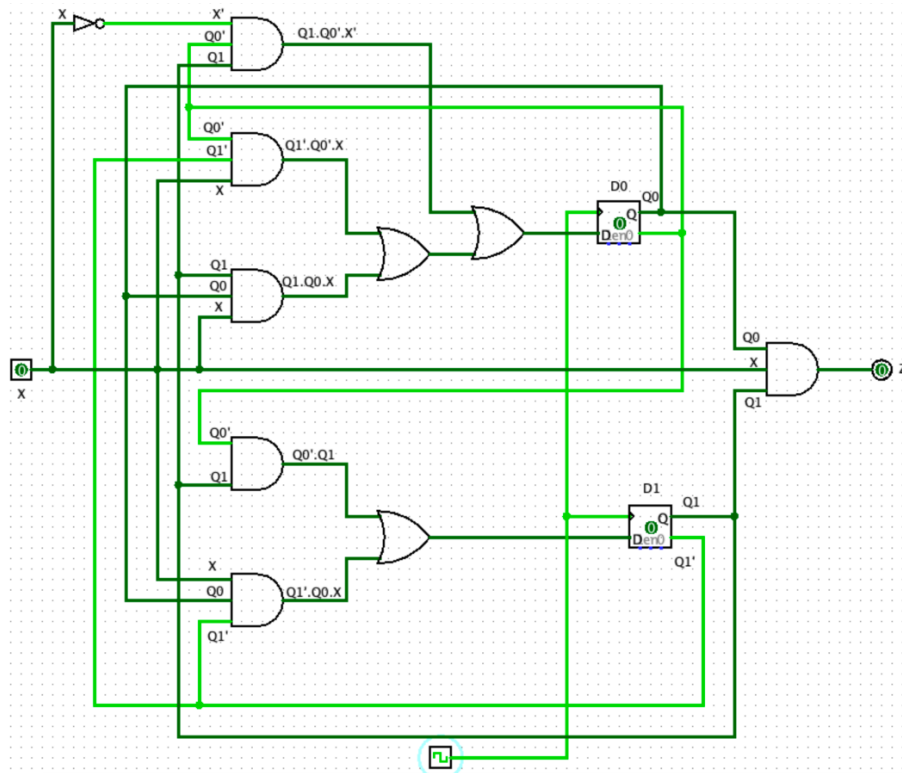


1.5 Circuit Simulation (LOGISIM snapshots)

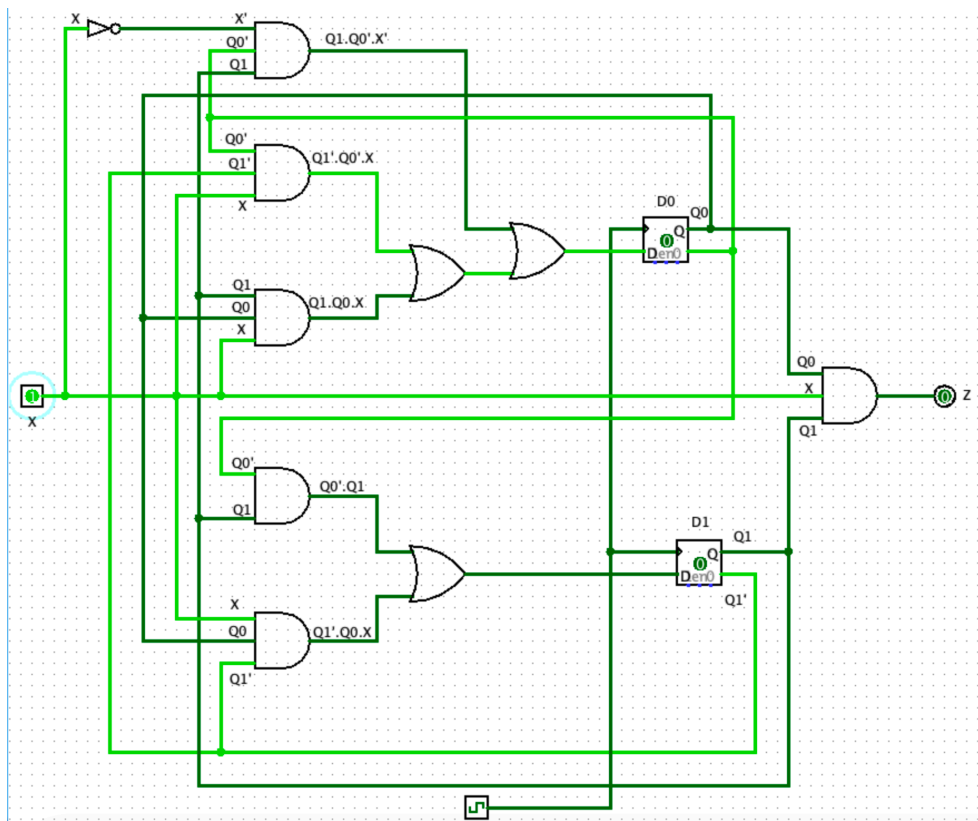
A1: Present state ($Q_1 = 0, Q_0 = 0, X = 0, Z = 0$)



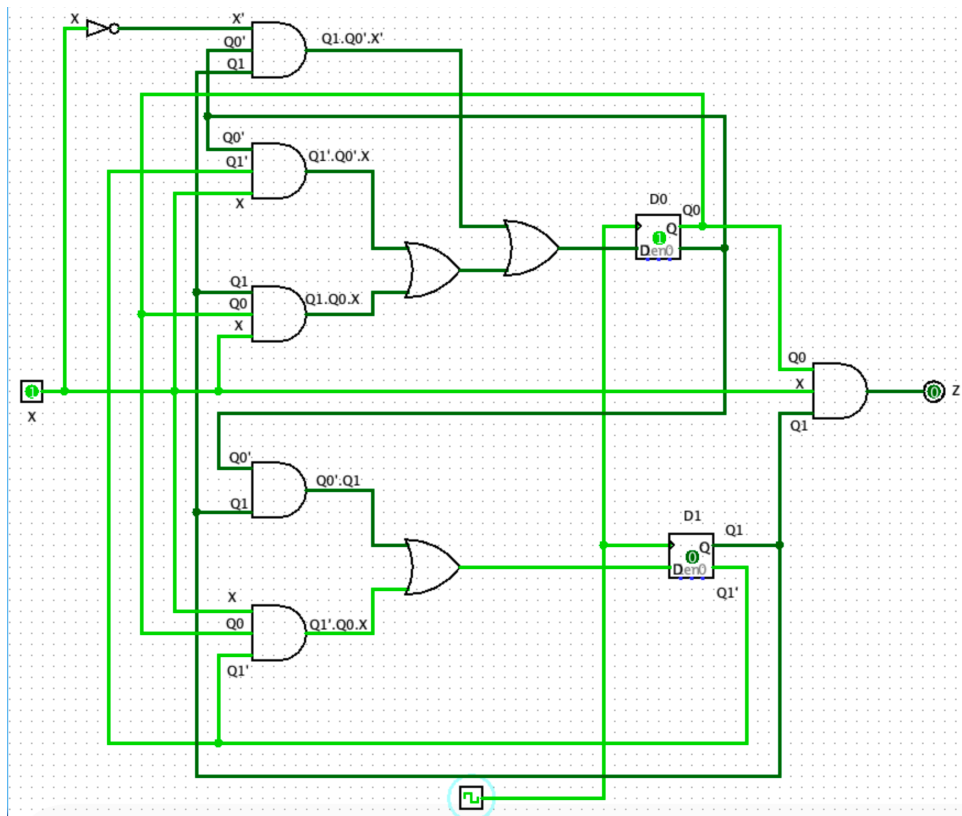
A2: Next state ($Q_1 = 0, Q_0 = 0, Z = 0$)



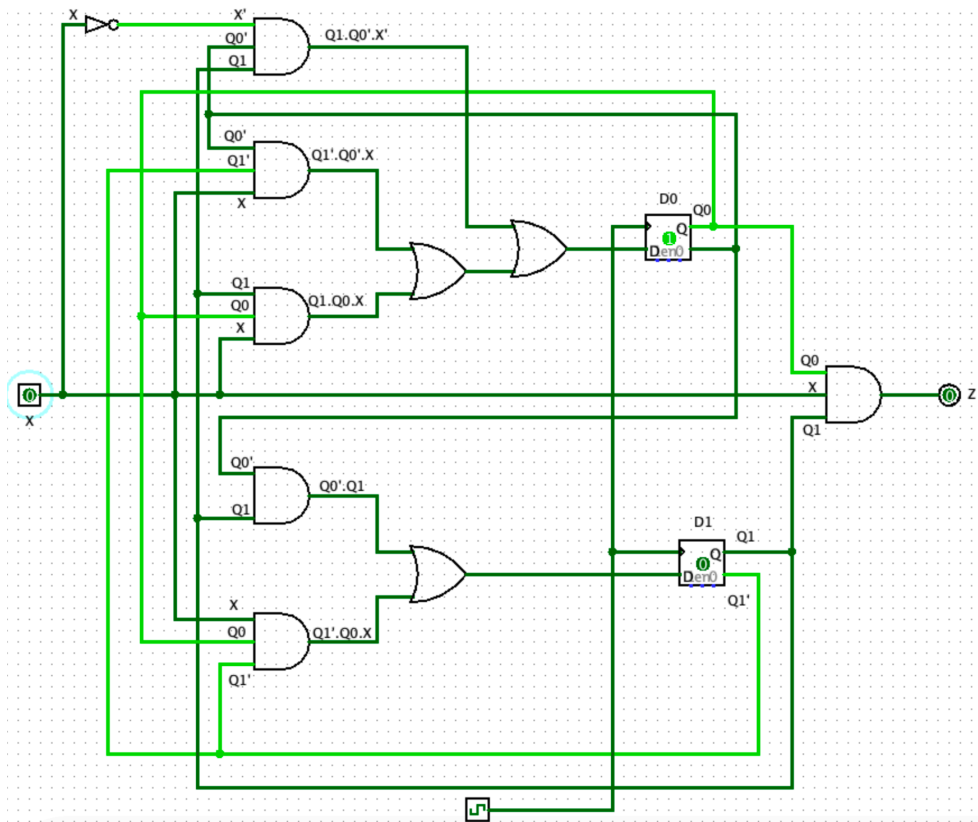
B1: Present state ($Q_1 = 0, Q_0 = 0, X = 1, Z = 0$)



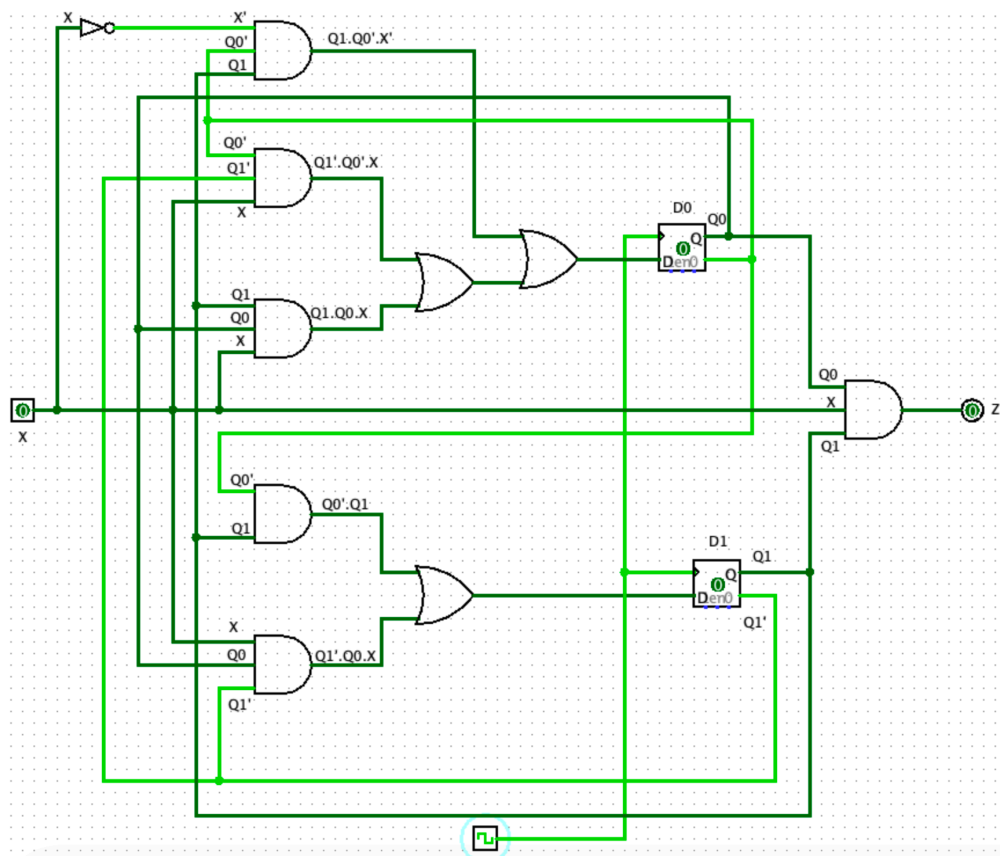
B2: Next state ($Q_1 = 0, Q_0 = 1, Z = 0$)



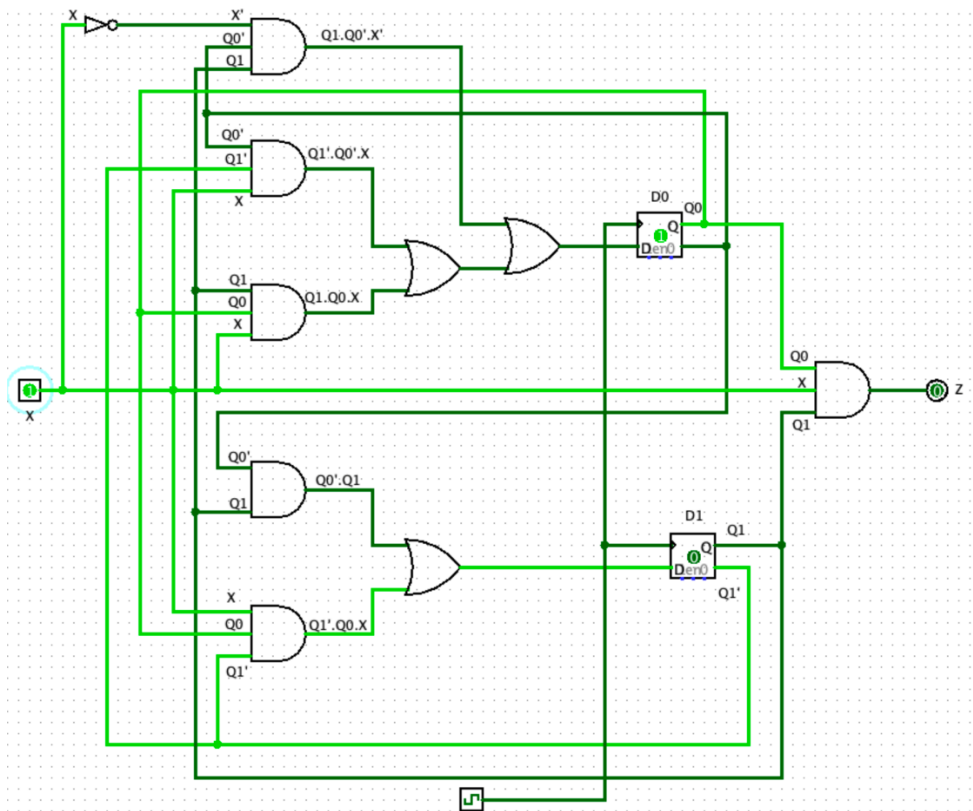
C1: Present state ($Q_1 = 0, Q_0 = 1, X = 0, Z = 0$)



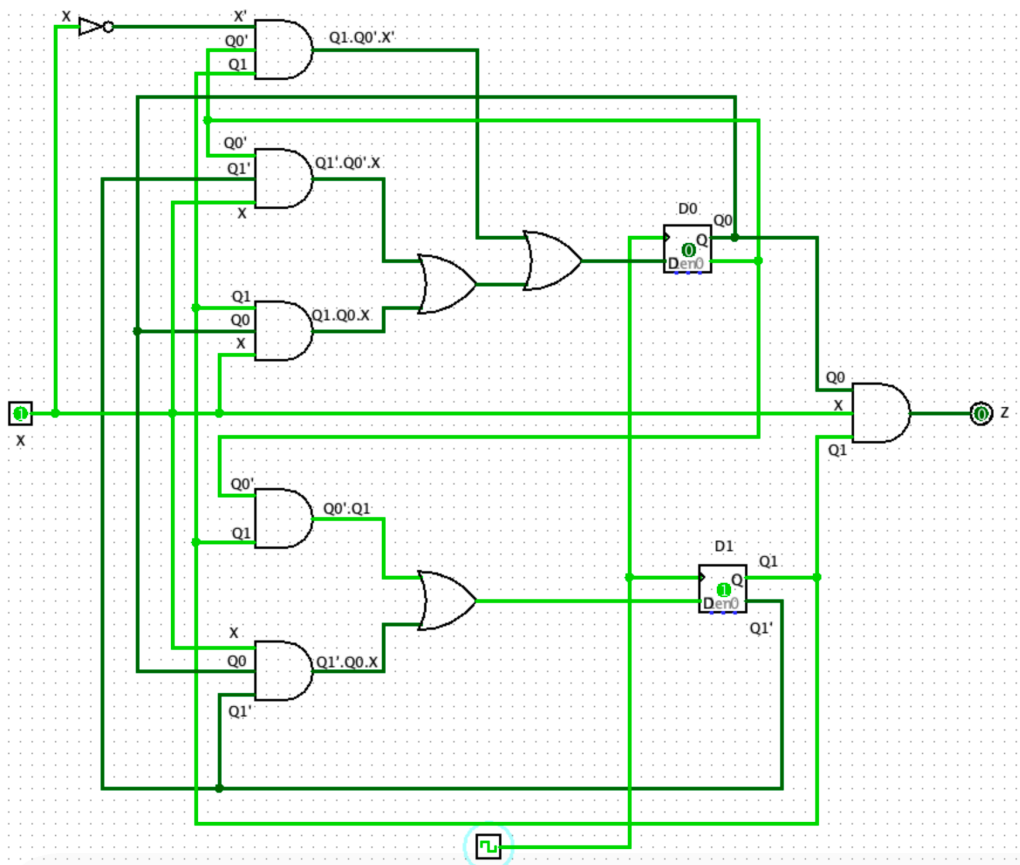
C2: Next state ($Q_1 = 0, Q_0 = 0, Z = 0$)



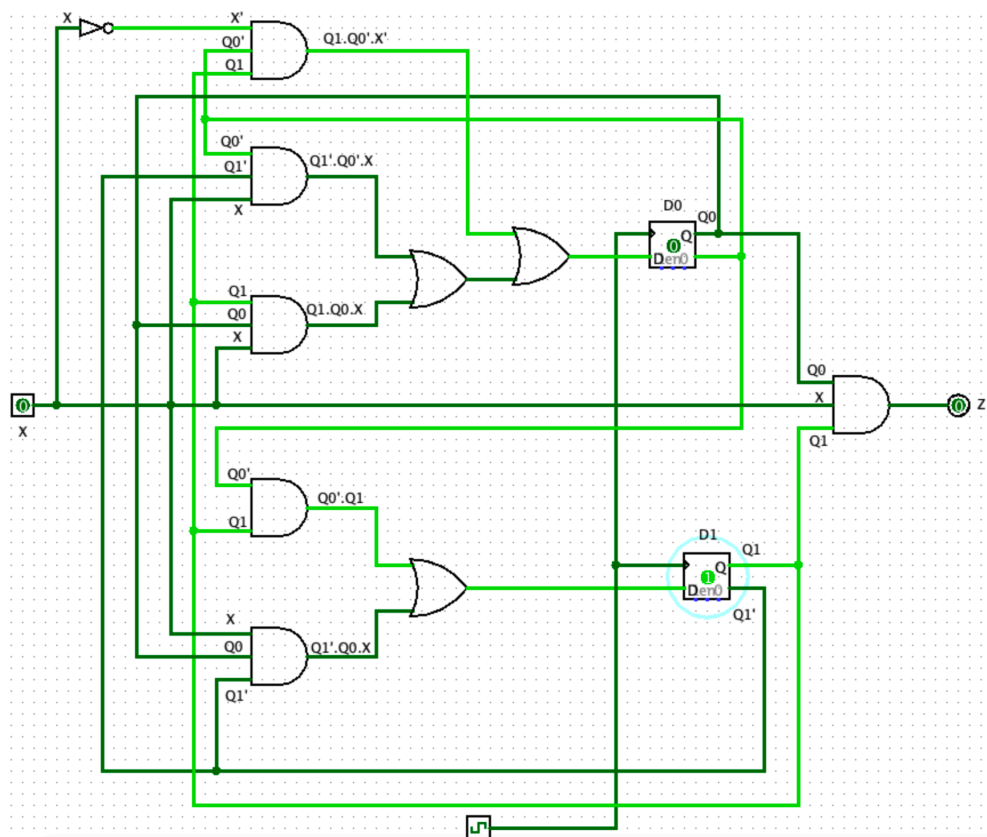
D1: Present state ($Q_1 = 0, Q_0 = 1, X = 1, Z = 0$)



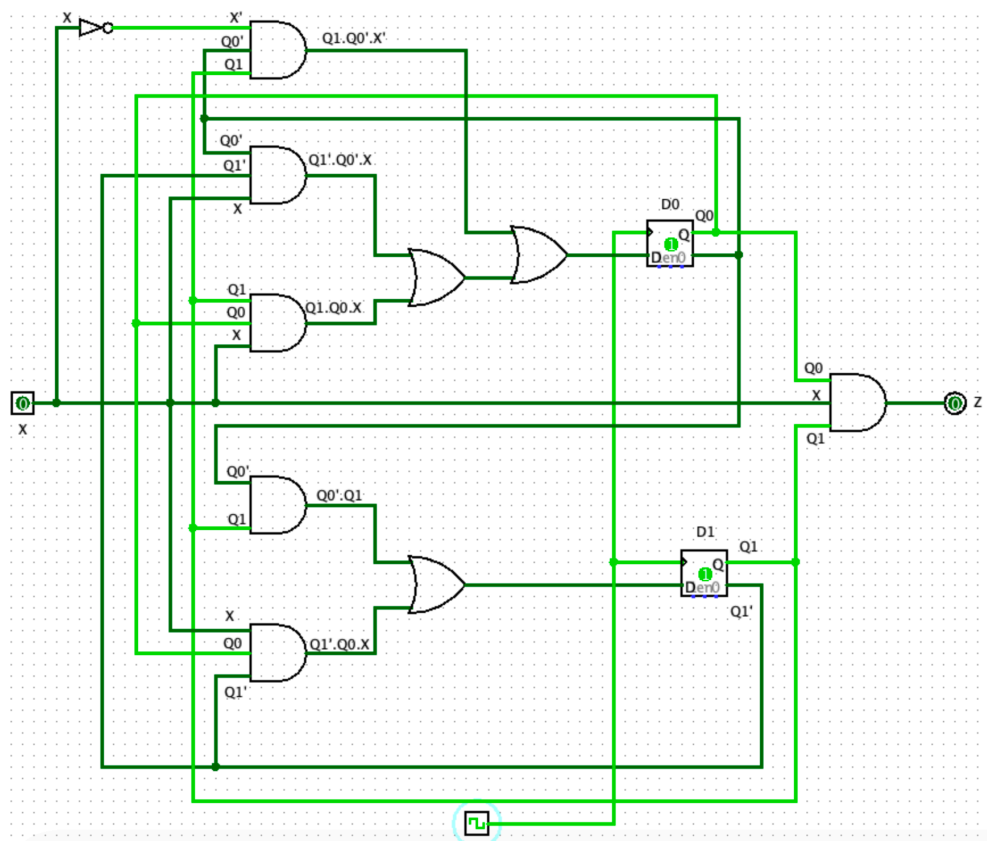
D2: Next state ($Q_1 = 1, Q_0 = 0, Z = 0$)



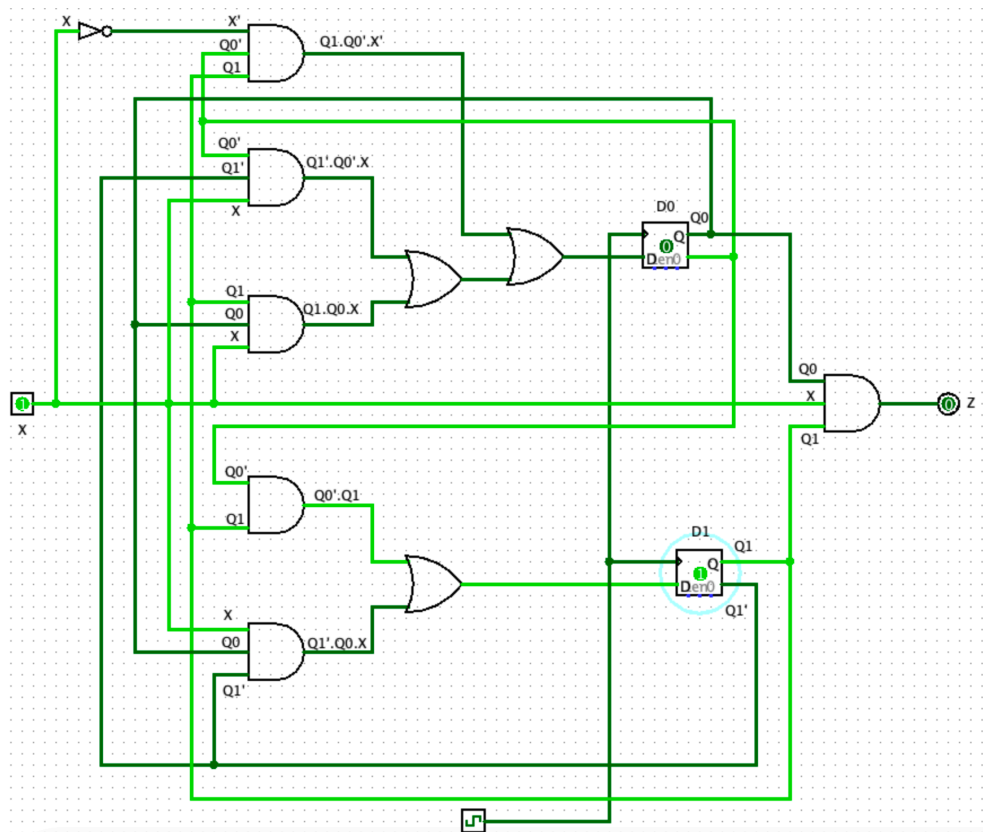
E1: Present state ($Q_1 = 1, Q_0 = 0, X = 0, Z = 0$)



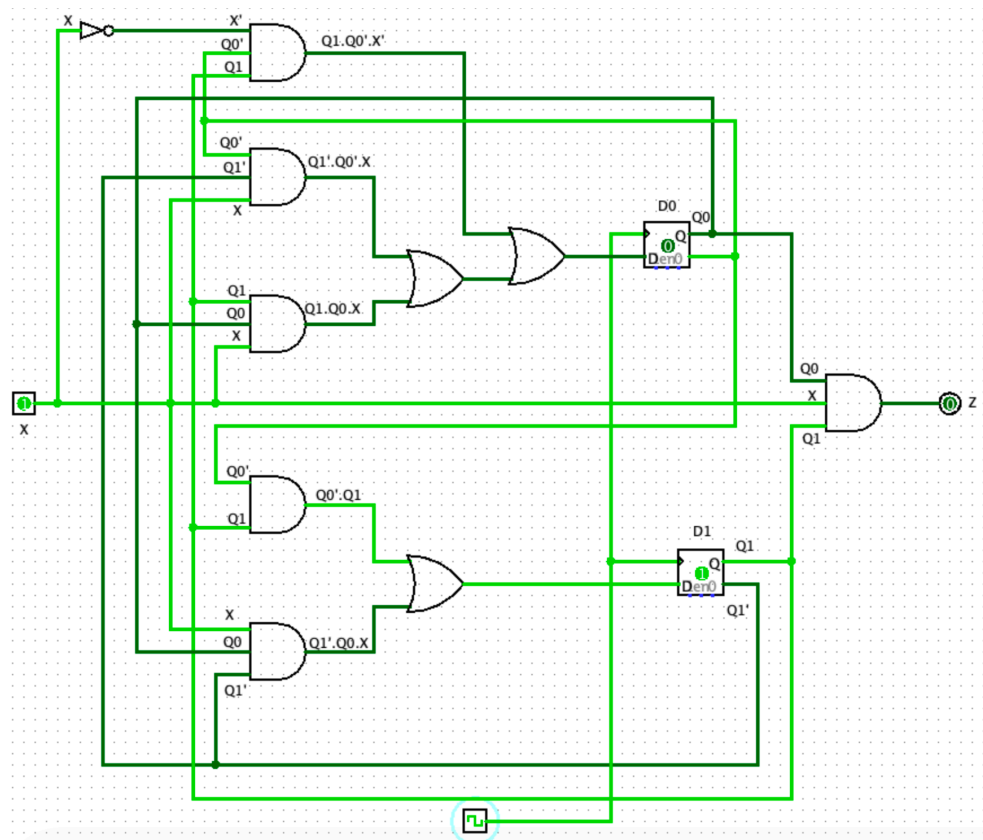
E2: Next state ($Q_1 = 1, Q_0 = 1, Z = 0$)



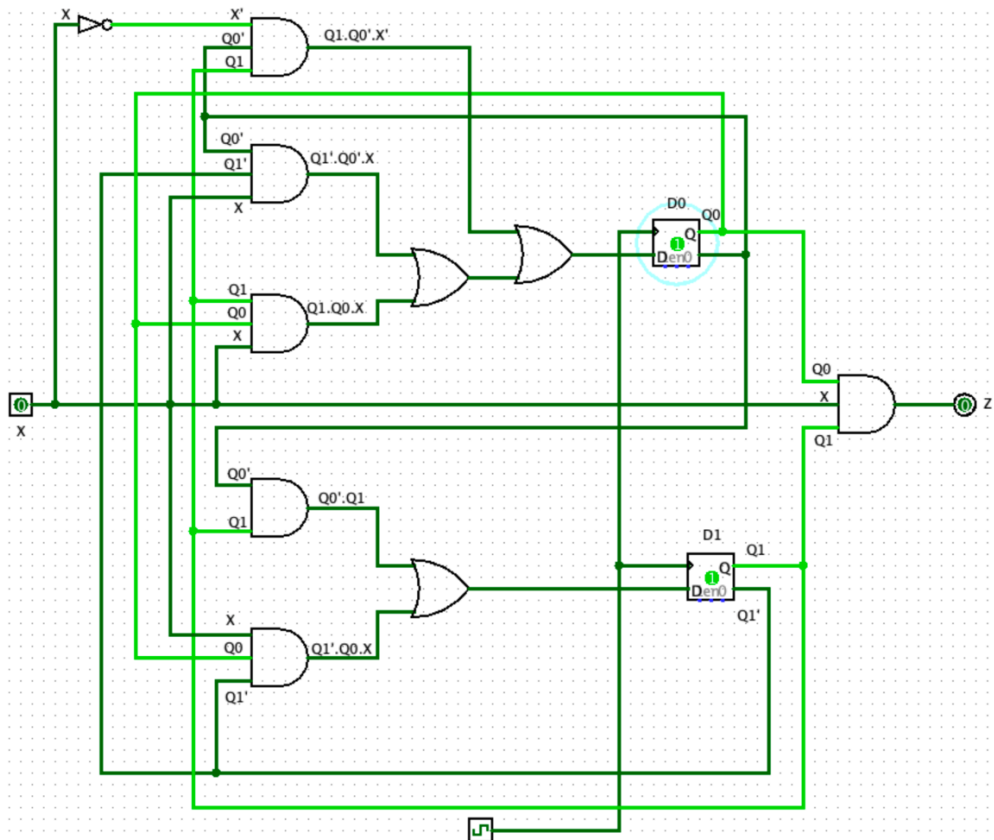
F1: Present state ($Q_1 = 1, Q_0 = 0, X = 1, Z = 0$)



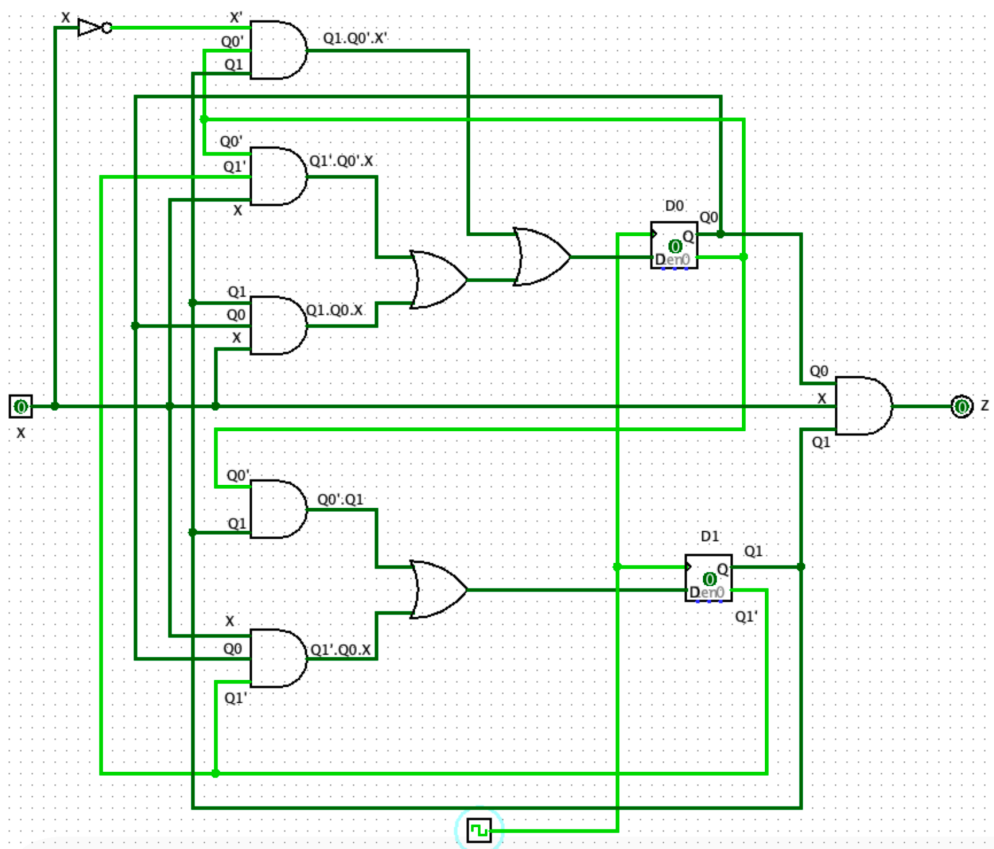
F2: Next state ($Q_1 = 1, Q_0 = 0, Z = 0$)



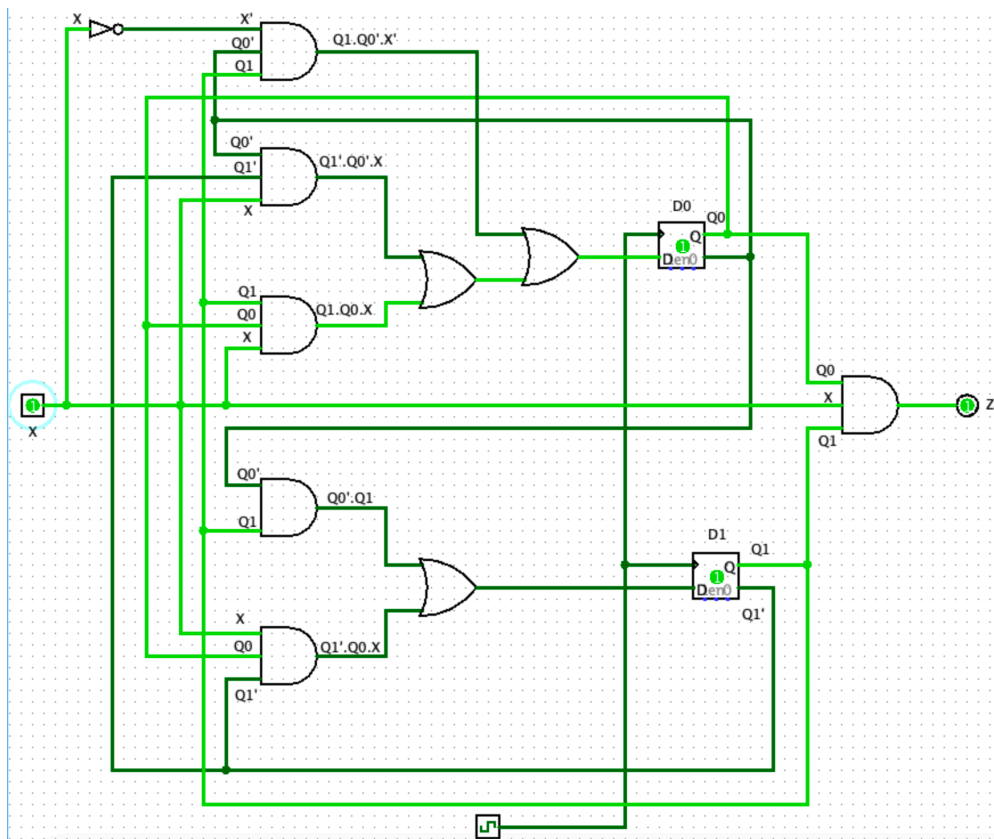
G1: Present state ($Q_1 = 1, Q_0 = 1, X = 0, Z = 0$)



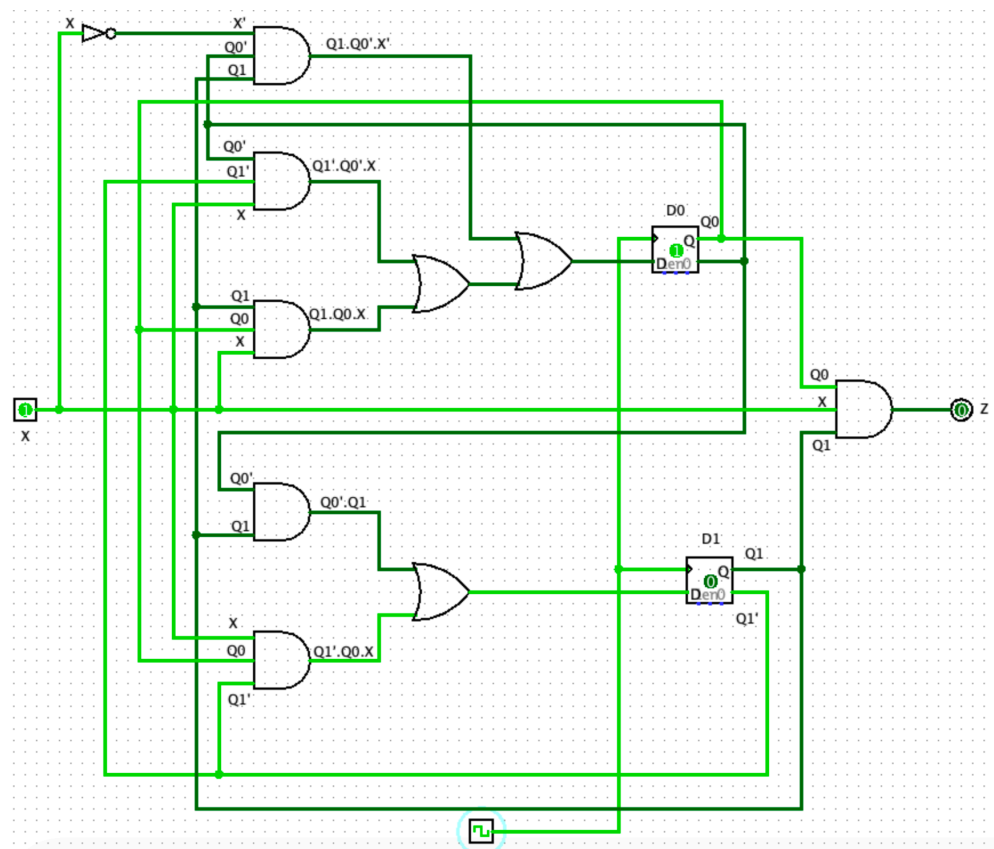
G2: Next state ($Q_1 = 0, Q_0 = 0, Z = 0$)



I1: Present state ($Q_1 = 1, Q_0 = 1, X = 1, Z = 1$)

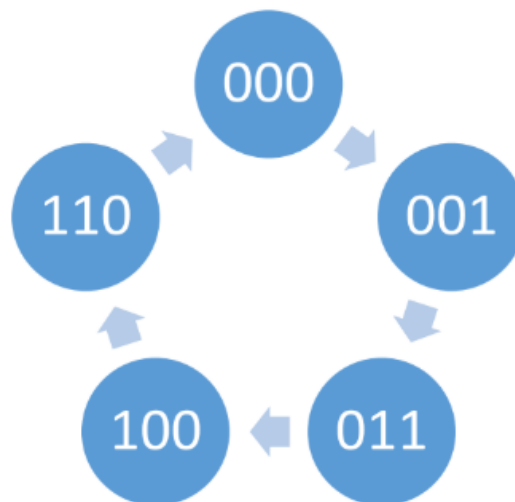


I2: Next state ($Q_1 = 0, Q_0 = 1, Z = 0$)



Question 2:

A synchronous counter counts the number of clock pulses received at its input. The synchronous counter uses the same clock signal from the same source and at exactly the same time. Using JK flip-flops, implement a synchronous counter with the following sequence:

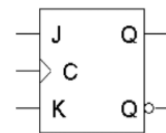
2.1 State Transition Diagram

2.2 Truth table

Present State			Next State			Flip-Flop Inputs					
Q2	Q1	Q0	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1	0	x	0	x	1	x
0	0	1	0	1	1	0	x	1	x	x	0
0	1	0	x	x	x	x	x	x	x	x	x
0	1	1	1	0	0	1	x	x	1	x	1
1	0	0	1	1	0	x	0	1	x	0	x
1	0	1	x	x	x	x	x	x	x	x	x
1	1	0	0	0	0	x	1	x	1	0	x
1	1	1	x	x	x	x	x	x	x	x	x

2.2.1 Excitation table for a JK flip-flop

Q(t)	Q(t+1)	J	K	Operation
0	0	0	x	No change / reset
0	1	1	x	Set / complement
1	0	x	1	Reset / complement
1	1	x	0	No change / set



2.3 Karnaugh maps

2.3.1 K-Map for J_2

Q_2Q_1 Q_0		$\bar{Q}_2\bar{Q}_1$ $\bar{Q}_2Q_1$ Q_2Q_1 $Q_2\bar{Q}_1$			
		00	01	11	10
$\bar{Q}_0$	0	0	X	X	X
Q_0	1	0	1	X	X

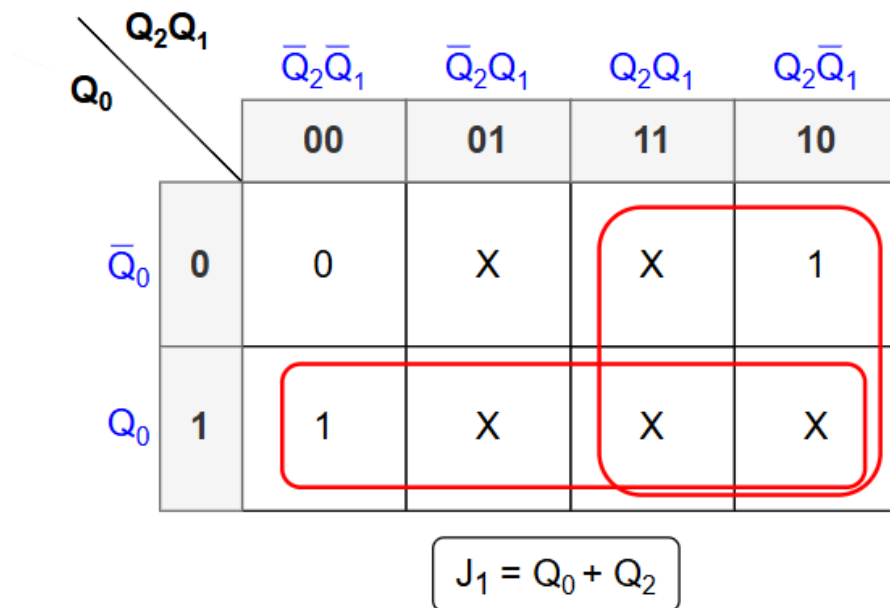
$J_2 = Q_1$

2.3.2 K-Map for K_2

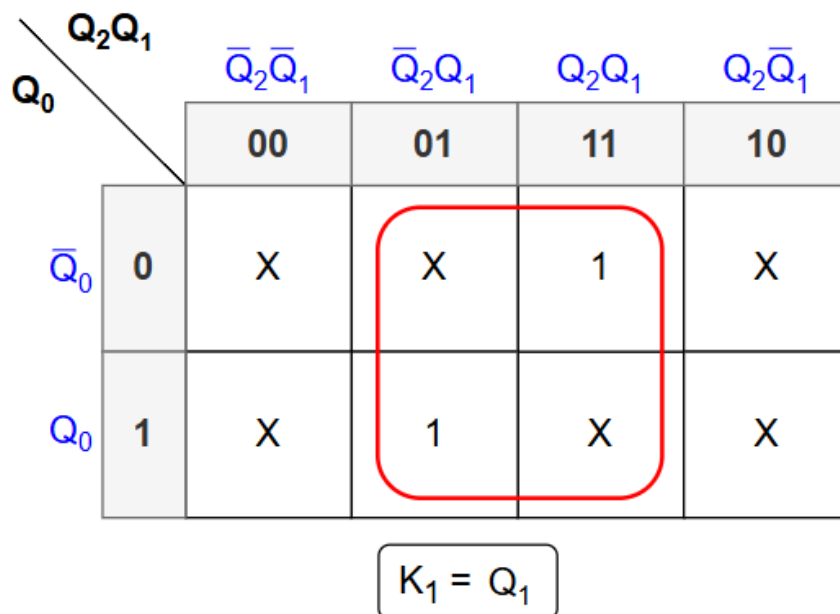
Q_2Q_1 Q_0		$\bar{Q}_2\bar{Q}_1$ $\bar{Q}_2Q_1$ Q_2Q_1 $Q_2\bar{Q}_1$			
		00	01	11	10
$\bar{Q}_0$	0	X	X	1	0
Q_0	1	X	X	X	X

$K_2 = Q_1$

2.3.3 K-Map for J_1



2.3.4 K-Map for K_1



2.3.5 K-Map for J_0

Q_2Q_1 Q_0		$\bar{Q}_2\bar{Q}_1$	$\bar{Q}_2Q_1$	Q_2Q_1	$Q_2\bar{Q}_1$
		00	01	11	10
$\bar{Q}_0$	0	1	X	0	0
Q_0	1	X	X	X	X

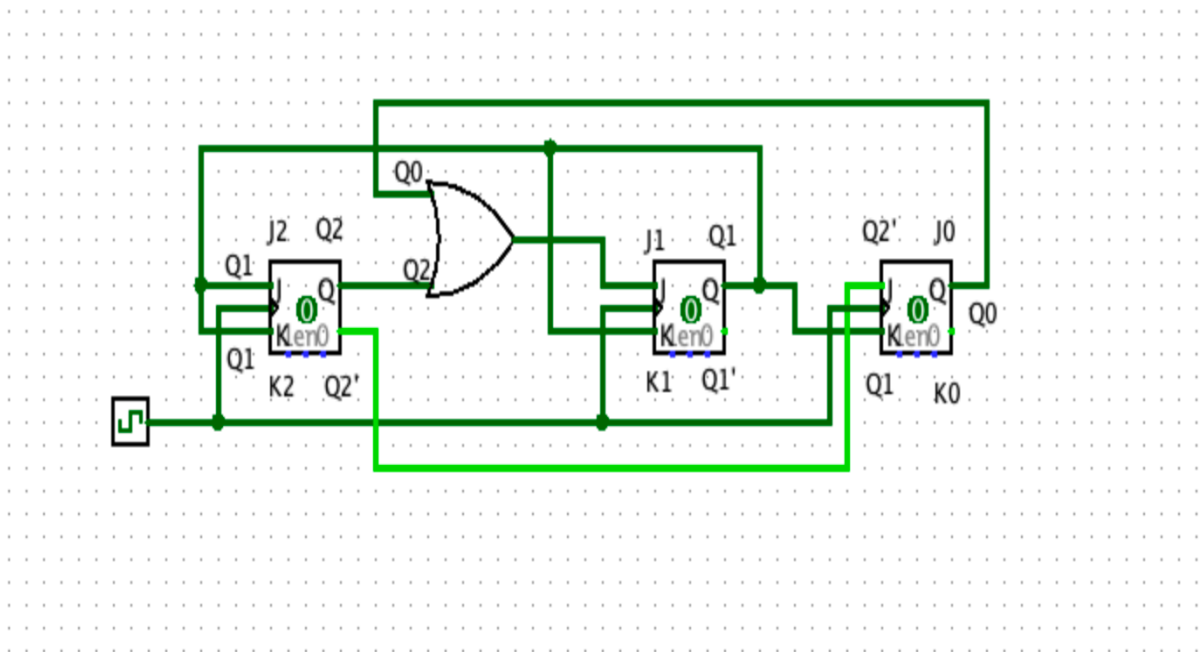
$J_0 = \bar{Q}_2$

2.3.6 K-Map for K_0

Q_2Q_1 Q_0		$\bar{Q}_2\bar{Q}_1$	$\bar{Q}_2Q_1$	Q_2Q_1	$Q_2\bar{Q}_1$
		00	01	11	10
$\bar{Q}_0$	0	X	X	X	X
Q_0	1	0	1	X	X

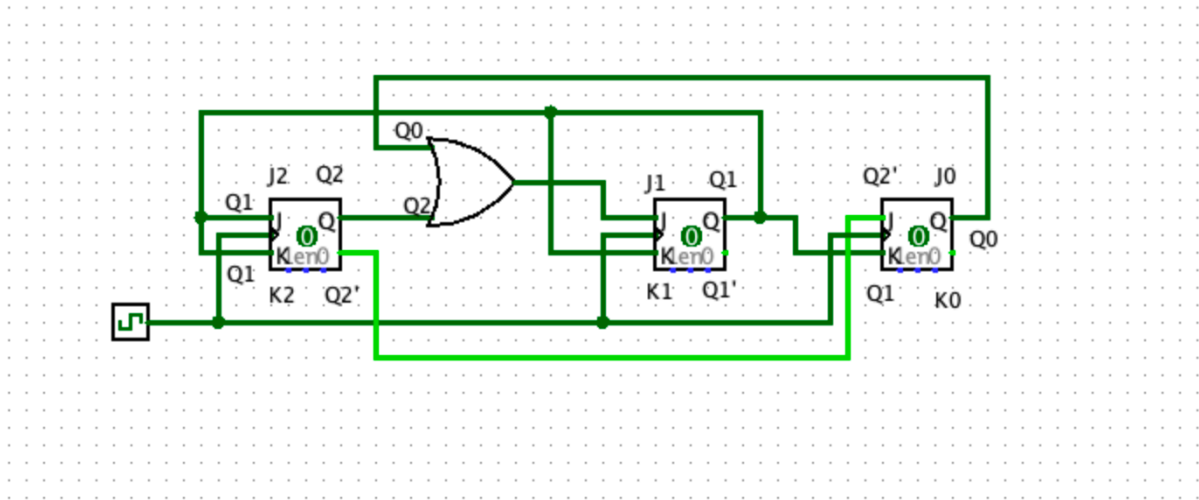
$K_0 = Q_1$

2.4 Main circuit

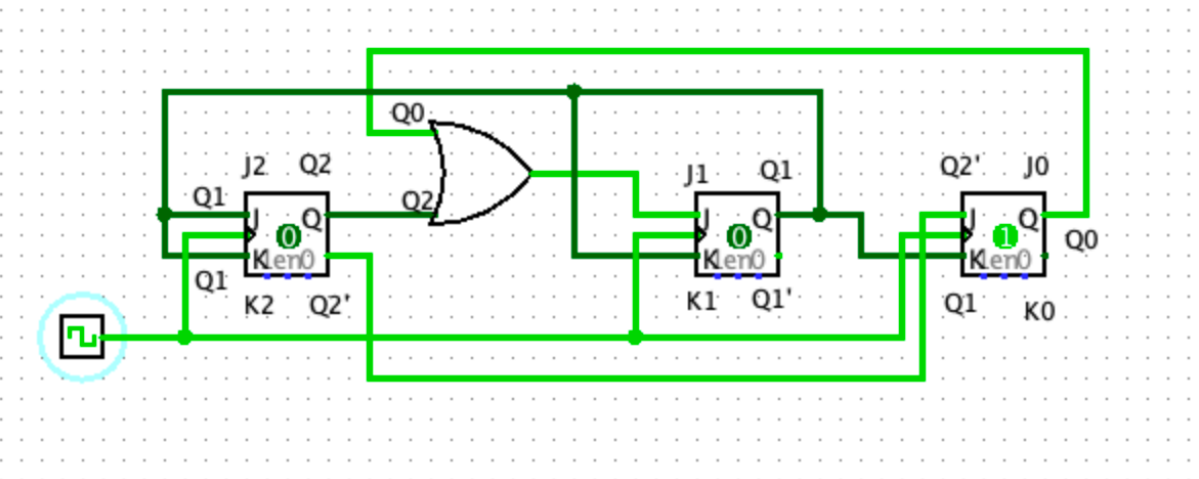


2.5 Circuit Simulation

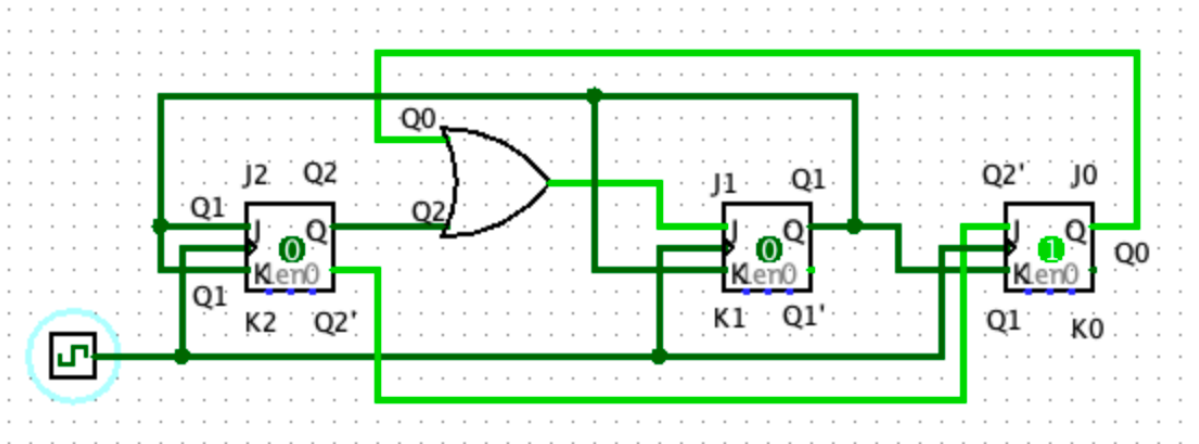
A1: Present state $Q_2 = 0$, $Q_1 = 0$, $Q_0 = 0$ (0 in decimal)



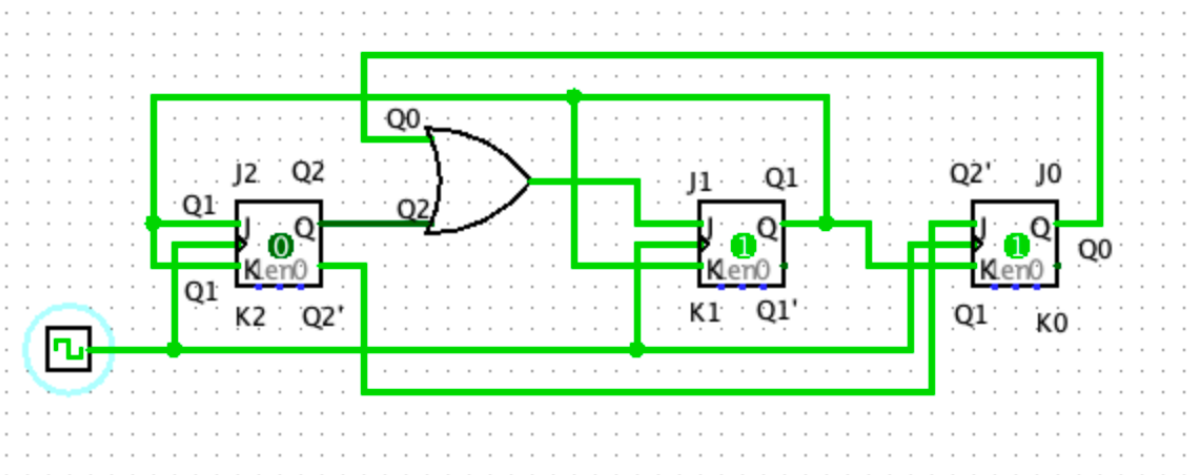
A2: Next state $Q_2 = 0$, $Q_1 = 0$, $Q_0 = 1$ (1 in decimal)



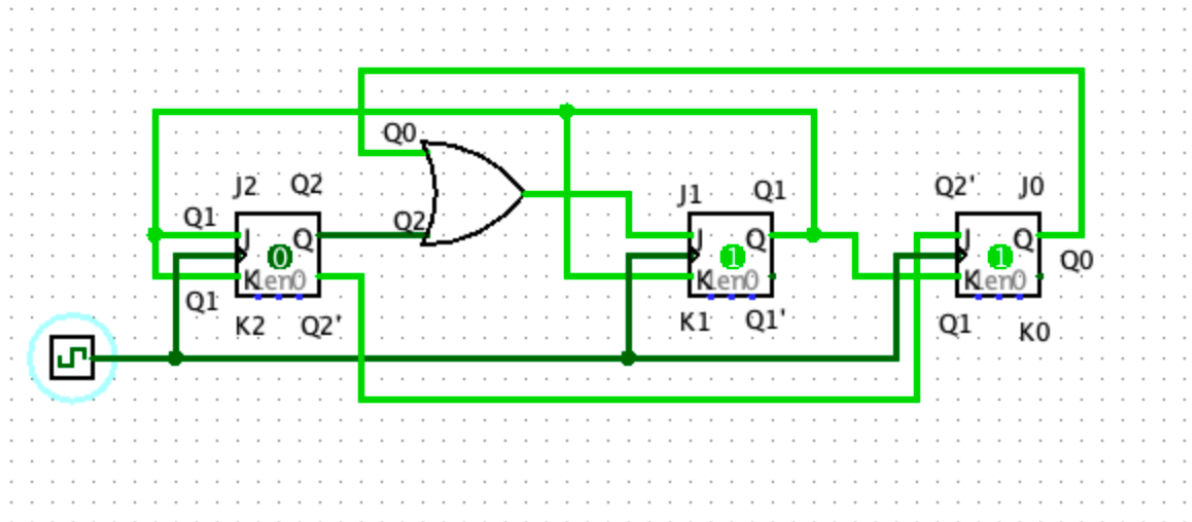
B1: Present state $Q_2 = 0, Q_1 = 0, Q_0 = 1$ (1 in decimal)



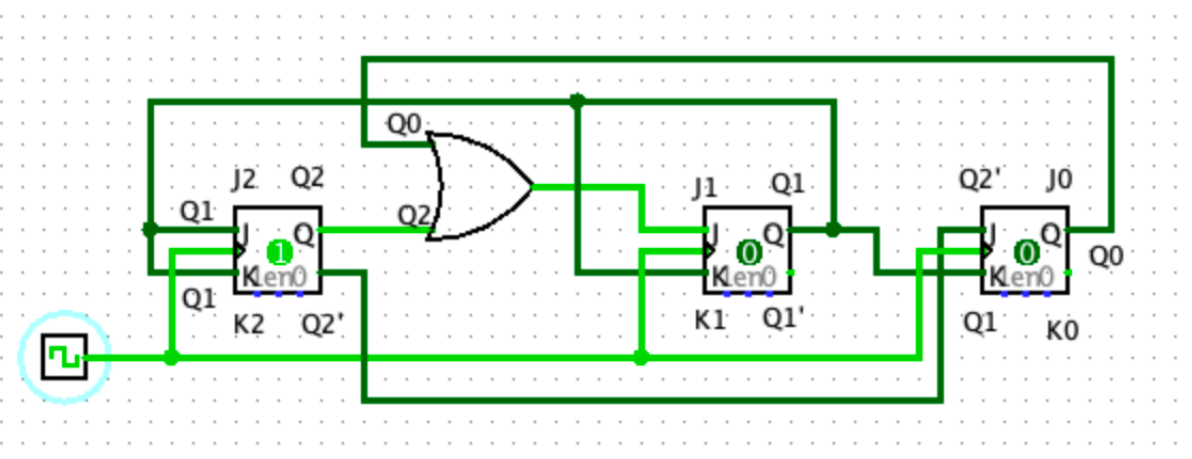
B2: Next state $Q_2 = 0, Q_1 = 1, Q_0 = 1$ (3 in decimal)



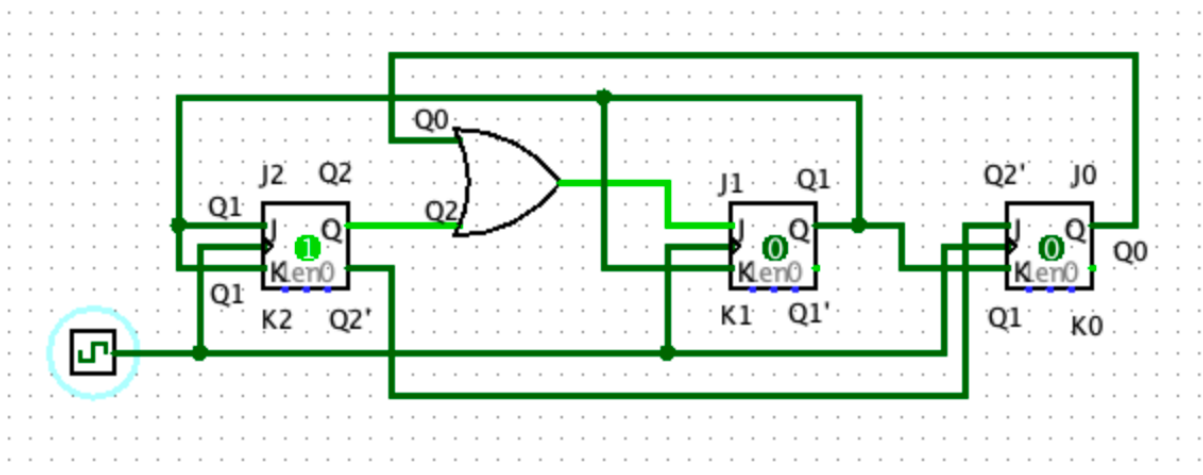
C1: Present state $Q_2 = 0, Q_1 = 1, Q_0 = 1$ (3 in decimal)



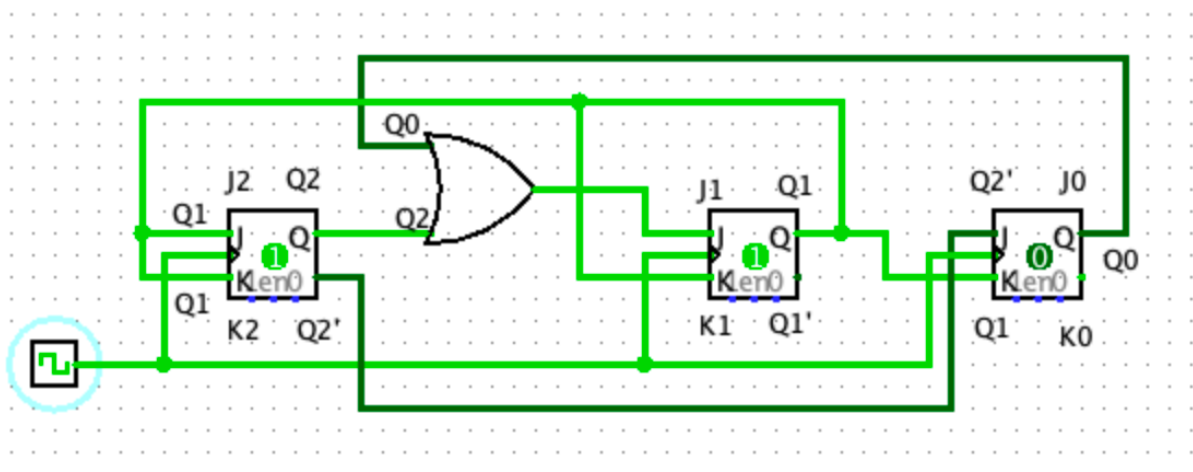
C2: Next state $Q_2 = 1, Q_1 = 0, Q_0 = 0$ (4 in decimal)



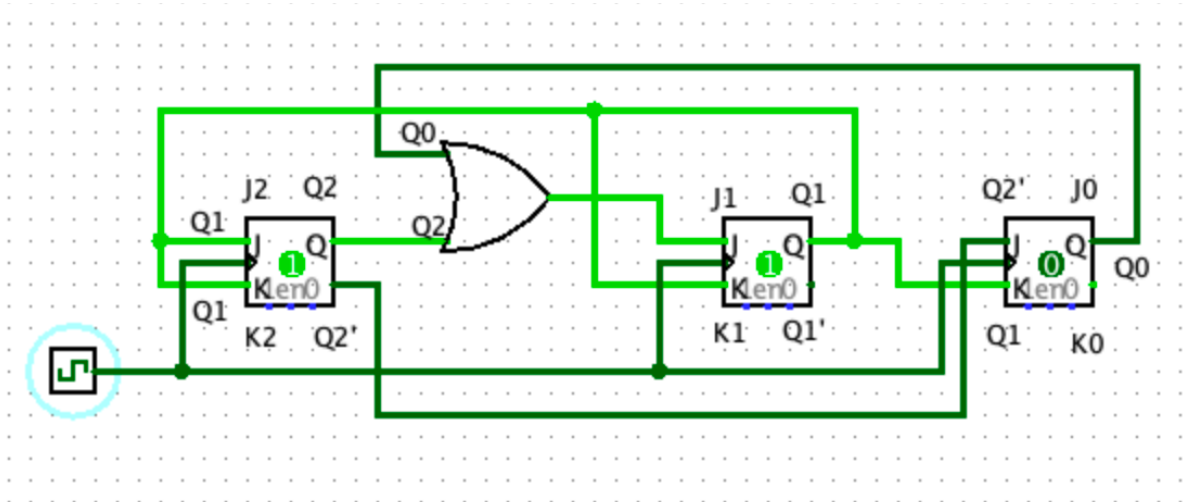
D1: Present state $Q_2 = 1, Q_1 = 0, Q_0 = 0$ (4 in decimal)



D2: Next state $Q_2=1, Q_1=1, Q_0 = 0$ (6 in decimal)



E1: Present state $Q_2 = 1, Q_1 = 1, Q_0 = 0$ (6 in decimal)



E2: Next state $Q_2 = 0, Q_1 = 0, Q_0 = 0$ (0 in decimal)

